

1. Executive Summary

Metric	Value
Report Period	2026-08-28 - 2026-09-04
Total Threats Mitigated	81
High Priority	25
Low Priority	56
Manual Releases	0
Manual Blocks	0
ML Metrics	Value
False Positives	159
FP Rate	0.12%
IF Precision	99.74%
IF Recall	91.09%
IF F1-Score	95.22%
IF Accuracy	96.86%

Traffic Summary	Count
Total Flows Observed	194945
Total Packets Analyzed	85974235199
Malicious Packets	85945357550
True Negatives Passed	133643
Attack Vector	Count
ICMP Flood	16
SYN Flood	21
UDP Flood	26
Uncertain	18
Action	Count
Quarantined	13
Time Ban	10
Blackhole	9
Released	15

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.74%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	91.09%	Share of actual attacks successfully flagged
F1-Score	95.22%	Balanced measure combining Precision and Recall
Accuracy	96.86%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.12%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	8.91%	Actual attacks that went undetected
True Positive Rate (TPR)	91.09%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.88%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 60914	FN 5961
Actual: Normal	FP 159	TN 127682

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	29299	0	0
Act: ICMP	0	15224	0
Act: UDP	0	0	16391

2c. Response Latency

Metric	Value	Description
Detection Time	558.2 ms	Average time taken to detect an attack
Mitigation Response Time	4.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.24%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	3.14%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.82%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

No offence history for this period.

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 12:59:48	10.0.0.6	DDoS	UDP Flood	90.6%	High	Time Ban (1m)
2026-09-04 12:59:48	10.0.0.8	DDoS	UDP Flood	90.9%	High	Time Ban (1m)
2026-09-04 12:59:56	10.0.0.5	DDoS	Uncertain	59.6%	Low	Released
2026-09-04 12:59:56	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-04 12:59:56	10.0.0.7	DDoS	Uncertain	68.4%	Low	Released
2026-09-04 13:00:49	10.0.0.6	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:00:49	10.0.0.8	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:01:11	10.0.0.8	DDoS	UDP Flood	90.9%	High	Quarantined
2026-09-04 13:02:26	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:27	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:30	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:32	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	90.2%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-04 13:03:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-04 13:03:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-04 13:03:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-04 13:03:39	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Released
2026-09-04 13:03:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (2m)
2026-09-04 13:05:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (5m)
2026-09-04 13:05:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (5m)
2026-09-04 13:05:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-04 13:05:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (5m)
2026-09-04 13:09:52	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-04 13:09:56	10.0.0.3	DDoS	Uncertain	59.1%	Low	Released
2026-09-04 13:09:57	10.0.0.1	DDoS	Uncertain	58.6%	Low	Released
2026-09-04 13:10:53	10.0.0.9	DDoS	UDP Flood	92.4%	High	Released
2026-09-04 13:11:25	10.0.0.6	DDoS	UDP Flood	73.6%	High	Quarantined
2026-09-04 13:12:26	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:12:27	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:12:29	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 13:12:47	10.0.0.23	DDoS	ICMP Flood	92.0%	High	Time Ban (1m)
2026-09-04 13:12:48	10.0.0.25	DDoS	ICMP Flood	91.9%	High	Time Ban (1m)
2026-09-04 13:12:48	10.0.0.23	DDoS	ICMP Flood	92.0%	Critical	Time Ban
2026-09-04 13:12:48	10.0.0.25	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-09-04 13:12:50	10.0.0.24	DDoS	ICMP Flood	92.0%	High	Time Ban (1m)
2026-09-04 13:12:50	10.0.0.24	DDoS	ICMP Flood	92.0%	Critical	Time Ban
2026-09-04 13:13:47	10.0.0.23	DDoS	ICMP Flood	92.3%	High	Time Ban (2m)
2026-09-04 13:13:48	10.0.0.25	DDoS	ICMP Flood	92.3%	High	Time Ban (2m)
2026-09-04 13:13:50	10.0.0.24	DDoS	ICMP Flood	92.2%	High	Released
2026-09-04 13:13:56	10.0.0.24	DDoS	ICMP Flood	92.2%	Critical	Time Ban (2m)
2026-09-04 13:15:47	10.0.0.23	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:15:48	10.0.0.25	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:15:56	10.0.0.24	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:19:58	10.0.0.2	DDoS	Uncertain	58.9%	Low	Released
2026-09-04 13:21:26	10.0.0.9	DDoS	Uncertain	48.4%	Low	Quarantined
2026-09-04 13:22:26	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:27	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:28	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:46	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban (1m)
2026-09-04 13:22:46	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban
2026-09-04 13:22:47	10.0.0.21	DDoS	UDP Flood	78.0%	High	Time Ban (1m)
2026-09-04 13:22:48	10.0.0.21	DDoS	UDP Flood	78.0%	High	Time Ban
2026-09-04 13:22:48	10.0.0.22	DDoS	UDP Flood	78.3%	High	Time Ban (1m)
2026-09-04 13:22:49	10.0.0.22	DDoS	UDP Flood	78.3%	High	Time Ban
2026-09-04 13:23:46	10.0.0.20	DDoS	UDP Flood	86.9%	High	Released
2026-09-04 13:23:47	10.0.0.21	DDoS	UDP Flood	78.0%	High	Released
2026-09-04 13:23:48	10.0.0.22	DDoS	UDP Flood	85.2%	High	Released
2026-09-04 13:23:52	10.0.0.20	DDoS	UDP Flood	86.9%	Critical	Time Ban (2m)
2026-09-04 13:23:54	10.0.0.22	DDoS	UDP Flood	85.2%	Critical	Time Ban (2m)
2026-09-04 13:23:54	10.0.0.21	DDoS	UDP Flood	79.7%	Critical	Time Ban (2m)
2026-09-04 13:25:55	10.0.0.22	DDoS	UDP Flood	85.8%	High	Time Ban (5m)
2026-09-04 13:25:55	10.0.0.21	DDoS	UDP Flood	79.7%	High	Time Ban (5m)
2026-09-04 13:25:59	10.0.0.20	DDoS	UDP Flood	86.9%	Critical	Time Ban (5m)
2026-09-04 13:33:00	10.0.0.21	DDoS	UDP Flood	79.0%	Critical	Time Ban (10m)
2026-09-04 13:33:36	10.0.0.17	DDoS	SYN Flood	93.6%	Critical	Blackhole
2026-09-04 13:34:07	10.0.0.24	DDoS	ICMP Flood	93.4%	Critical	Blackhole
2026-09-04 13:35:33	10.0.0.19	DDoS	SYN Flood	91.9%	Critical	Blackhole
2026-09-04 13:35:33	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Blackhole

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 13:35:35	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Blackhole
2026-09-04 13:35:47	10.0.0.20	DDoS	UDP Flood	87.3%	Critical	Blackhole
2026-09-04 13:36:04	10.0.0.23	DDoS	ICMP Flood	96.3%	Critical	Blackhole
2026-09-04 13:36:06	10.0.0.25	DDoS	ICMP Flood	93.8%	Critical	Blackhole
2026-09-04 13:44:43	10.0.0.22	DDoS	UDP Flood	76.6%	Critical	Blackhole
2026-09-04 13:44:56	10.0.0.23	DDoS	Uncertain	-	Low	Released

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE